

90/100

CHE 312 HW 8

(9.15)

(a)

In winter, a frost line appears at the bottom of window rather than top because boundary layer is thinnest at the top of the window so heat flux is maximum. On other hand at bottom, the boundary layer is thickest resulting in formation of frost. Also air at the bottom is cooler ~~than~~ because it is denser than hot air.

(b)

Heat loss from room to the window by convection & radiation is,

$$q_{\text{loss}} = q_{\text{conv}} + q_{\text{rad}} \quad \text{--- (1)}$$

$$q_{\text{loss}} = A_s [h_L (T_w - T_s) + \epsilon \sigma (T_{\text{sur}}^4 - T_s^4)] \quad \text{--- (2)}$$

Using Churchill correlation,

$$\overline{Nu}_L = \frac{h_L L}{k} = \left(0.825 + \frac{0.387 Ra_L^{1/4}}{[1 + (0.498/Pr)^{1/4}]^{1/4}} \right)^2 \quad \text{--- (3)}$$

$$Ra_L = g \beta T (T_w - T_s) L^3 / \nu \alpha \quad \text{--- (4)}$$

$$Ra_L = 1.084 \times 10^{10} \quad \overline{Nu}_L = 258.9 \quad h_L = 3.6 \text{ W/m}^2\text{K}$$

heat loss with $\epsilon = 5.67 \times 10^{-8} \text{ W/m}^2\text{K}^4$ is

$$q_{\text{loss}} = (1 \times 1.8) [3.6(15) + 0.94 \epsilon (988^4 - 273^4)]$$

$$q_{\text{loss}} = 96.1 + 127.1$$

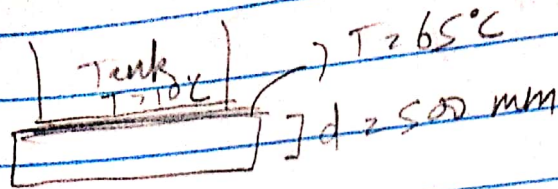
$$q_{\text{loss}} = 223 \text{ W}$$

Cost, $q_{\text{loss}} \times \text{utility rate} \times 24 \text{ hr}$

$$223 \text{ W} \times 10^{-3} \frac{\text{kW}}{\text{W}} \times 0.18 \frac{\$}{\text{kWh}} \times 24 \frac{\text{hr}}{\text{day}}$$

$$= 0.963 \$/\text{day}$$

(9.28)



Temp of oil $= 10^\circ\text{C}$
Temp of heat surfaces $= 65^\circ\text{C}$

Property of oil: $\nu = 288 \times 10^{-6} \text{ m}^2/\text{s}$
 $k = 0.1445 \text{ W/mK}$
 $\alpha = 0.847 \times 10^{-7}$
 $\beta = 0.7 \times 10^{-3}/\text{K}$

$$L = \frac{A_s}{\beta} = \frac{\pi D^2/4}{\pi D} = \frac{D}{4} = 0.125$$

$$\begin{aligned} Ra_L &= \frac{g \beta (T_s - T_\infty) L^3}{\nu \alpha} \\ &= \frac{(9.8)(0.7 \times 10^{-3})(65 - 10)(0.125)^3}{288 \times 10^{-6} \times 0.847 \times 10^{-7}} \\ &= 3.02 \times 10^7 \end{aligned}$$

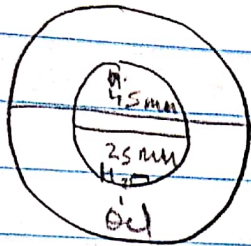
Heat rate from bottom heater surface to oil

$$\begin{aligned} Nu_L &= 0.15 Ra_L^{1/3} \\ &= (0.15) \times (3.02 \times 10^7)^{1/3} \\ &= 46.711 \end{aligned}$$

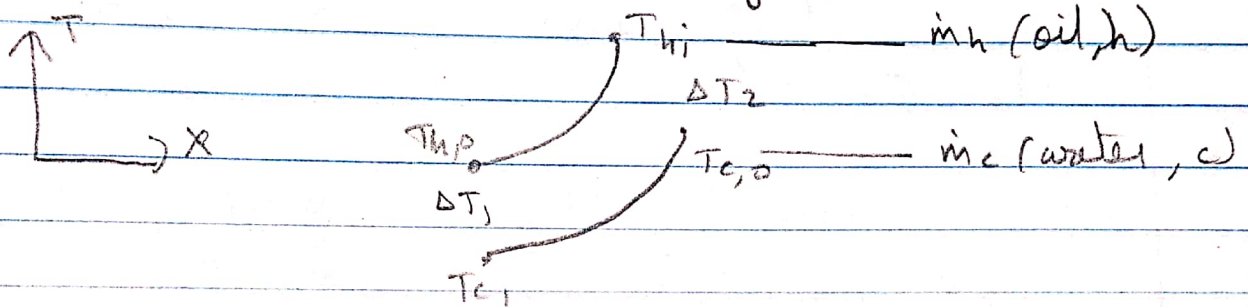
$$Nu_L = \frac{h_c L}{k}$$

$$h_c = \frac{Nu_L k}{L} = \frac{46.711 \times 0.1445}{0.125} = 54.18 \text{ W/m}^2\text{K}$$

$$\begin{aligned} q &= h_c A_s (T_s - T_\infty) \\ &= (54.18) \times \pi (0.5)^2 (65 - 10) = 588.8 \text{ W} \end{aligned}$$



Counter flow heat exchanger,



Property of hot fluid:

$$\begin{aligned} \rho &= 800 \text{ kg/m}^3 \\ C_{p,h} &= 1.9 \text{ kJ/kg}\cdot\text{K} \\ \nu_h &= 1 \times 10^{-5} \text{ m}^2/\text{s} \\ k_{f,h} &= 134 \times 10^{-3} \text{ W/m}\cdot\text{K} \\ Pr_{f,h} &= 140 \end{aligned}$$

Property of cold fluid:

$$\begin{aligned} \rho &= 1000 \text{ kg/m}^3 \\ C_{p,c} &= 4.2 \text{ kJ/kg}\cdot\text{K} \\ \nu_c &= 7 \times 10^{-7} \text{ m}^2/\text{s} \\ k_{f,c} &= 640 \times 10^{-3} \text{ W/m}\cdot\text{K} \\ Pr_{f,c} &= 4.7 \end{aligned}$$

(a) $q = \dot{m}_h C_h (T_{h,i} - T_{h,o})$

Hot = 50°C $T_{c,w} = ?$

$\dot{m} = 0.1 \text{ kg/s}$

$C_h = 1900 \text{ J/kg}\cdot\text{K}$

$T_{h,i} = 100, T_{h,o} = 60$

$$\dot{m}_h C_p (T_{h,i} - T_{h,o}) = \dot{m}_c C_p (T_{c,o} - T_{c,i})$$

$$C_p (T_h - T_o) = C_p (T_{c,o} - T_i)$$

$$T_{c,o} = \frac{9500 + 1272600}{4200} = 325.62 \text{ K}$$

(b)

L = 27

$$\begin{aligned} Q &= h A (T_m - T_o) \\ &= m c_p (T_{h,i} - T_{h,e}) \\ &= (0.1) (1900) (373 - 323) \\ &= 9500 \text{ W} \end{aligned}$$

$$U_o = 55 \text{ W/m}^2$$

$$Q = U_o A_o (\text{LMTD})$$

$$\text{LMTD} = \frac{\Delta T_1 - \Delta T_2}{\ln \frac{T_2}{T_1}}$$

$$\begin{aligned} \text{LMTD} &= \frac{(373 - 325.62) - (323 - 303)}{\ln \left(\frac{373 - 325.62}{323 - 303} \right)} \\ &= 31.95^\circ\text{C} \end{aligned} \quad \checkmark$$

$$Q = U_o (\pi D L) (\text{LMTD})$$

$$L = \frac{Q}{(\pi D) U_o (\text{LMTD})}$$

$$L = \frac{9500 \text{ W}}{\pi (0.025) (55) (31.27)}$$

$$L = 69.2 \text{ m} \quad \checkmark$$

The heat transfer rate

$$\epsilon = \frac{q}{q_{\max}}$$

$C_{\min} = \text{min heat capacity rate}$
 $= \dot{m}_{\text{air}} C_{\text{air}}$

$$q = \epsilon q_{\max}$$

$$= q_{\text{Main}}^{-0.2}$$

$$q = \dot{m}_{\text{air}} C_{\text{air}} (T_h - T_{c,i}) \times \text{Main}^{+0.2}$$

$$= \text{Main}^{0.8} C_{\text{air}} (T_h - T_{c,i})$$

$$q_{\text{medium}} = (\text{Main})^{0.8} C_{\text{air}} (T_h - T_{c,i})$$

$$q_{\text{high}} = (\text{Main})^{0.8} C_{\text{air}} (T_h - T_{c,i})$$

$$\therefore \frac{q_{\text{high}}}{q_{\text{medium}}} = \frac{(\text{Main})^{0.8} C_{\text{air}} (T_h - T_{c,i})}{(\text{Main})^{0.8} C_{\text{air}} (T_h - T_{c,i})}$$

$$(\dot{m}_{\text{air}})_{\text{high}} = 2(\dot{m}_{\text{air}})_{\text{medium}}$$

$$\frac{q_{\text{high}}}{q_{\text{medium}}} = 2^{0.8} = 1.7411$$

$$q_{\max} = 1.741 q_{\text{medium}}$$

$$\therefore \text{increase } q = \frac{q_{\max} - q_{\text{medium}}}{q_{\text{medium}}} = \frac{1.741 - 1}{1}$$

$\%$

$$= 0.741$$

$$= 74.1\%$$

68%

(b) when fan is turned to medium mode,

$$m_{\text{medium}} = 1.5 m_{\text{low}} \quad \text{--- (2)}$$

$$q_{\text{medium}} = 1.2 q_{\text{low}} \quad \text{--- (3)}$$

In low speed condition air $0 \rightarrow 30^\circ\text{C}$

$$q_{\text{medium}} = (m_{\text{air}})_{\text{medium}} C_{\text{air}} (T_h - T_{\text{ci}})$$

$$q_{\text{low}} = (m_{\text{air}})_{\text{low}} C_{\text{air}} (T_h - T_{\text{ci}})$$

$$\text{low} \rightarrow 30 \rightarrow 0$$

$$q_{\text{low}} = (m_{\text{air}})_{\text{low}} C_{\text{air}} (30 - 0)$$

$$\frac{q_{\text{med}}}{q_{\text{low}}} = \frac{(m_{\text{air}})_{\text{medium}} C_{\text{air}} (T_h - T_{\text{ci}})}{(m_{\text{air}})_{\text{low}} C_{\text{air}} (30 - 0)}$$

$$1.2 = 1.5 \frac{(T_h - T_{\text{ci}})}{30}$$

$$T_h = 24^\circ\text{C}$$

$$\times$$
$$26.1^\circ\text{C}$$